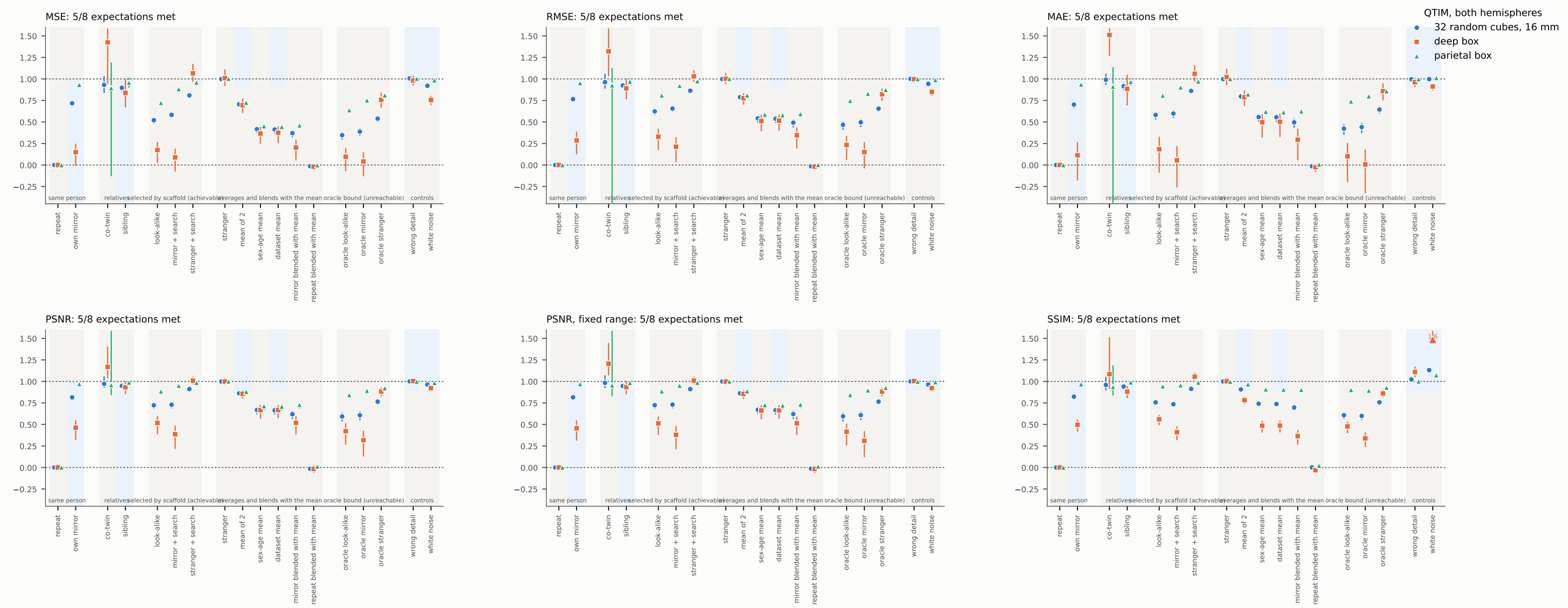


Expectations, random cubes (value = R or contrast)

	repeat	own mirror	co-twin	sibling	look-alike	mirror + search	stranger + search	stranger	mean of 2	sex-age mean	dataset mean	mirror blended with mean	repeat blended with mean	oracle look-alike	oracle mirror	oracle stranger	wrong detail	white noise
MSE	pass 0.72	pass 0.90	fail 0.71	fail 0.41	pass -0.01	fail -0.35	pass 1.01	pass 0.92										
RMSE	pass 0.77	pass 0.92	fail 0.79	fail 0.54	pass -0.01	fail -0.27	pass 1.00	pass 0.94										
MAE	pass 0.70	pass 0.92	fail 0.80	fail 0.55	pass -0.01	fail -0.21	pass 1.00	pass 1.00										
PSNR	pass 0.81	pass 0.95	fail 0.86	fail 0.66	pass -0.01	fail -0.20	pass 1.00	pass 0.96										
PSNR, fixed range	pass 0.82	pass 0.95	fail 0.86	fail 0.67	pass -0.01	fail -0.20	pass 1.00	pass 0.96										
SSIM	pass 0.82	pass 0.94	fail 0.91	fail 0.74	pass 0.00	fail -0.13	pass 1.02	pass 1.13										
Pearson r	pass 0.82	pass 0.91	fail 0.87	fail 0.58	pass -0.01	fail -0.26	pass 1.01	pass 0.99										
ACC vs dataset mean	pass 0.75	pass 0.90	pass 1.00	n/a	pass -0.00	pass -0.00	pass 1.00	pass 1.00										
ACC vs sex-age mean	pass 0.75	pass 0.90	pass 0.99	fail 0.89	pass -0.00	pass -0.02	pass 1.00	pass 1.00										
IDDT-like	pass 0.89	pass 0.96	fail 0.75	fail 0.38	pass -0.07	fail -0.50	pass 1.06	pass 2.16										
	E1 mirror<str		E2 sib<str		E3a mean2			E3b mean						E4a shrink rep		E4b shrink mir		E5a wrong		E5b noise

R = (d - d_repeat) / (d_stranger - d_repeat): 0 = repeat scan, 1 = stranger (mean of 16). Bars: 95% interval, bootstrap over families. Blue bands: where an identity-faithful metric is expected (mirror and sibling below 1; averages, wrong detail and noise at or above 0.9).
pre-QC: affine registration only, MRI registration gate not met (results/registration/status.md)